Multispectral Scene Classification via Cross-Modal Knowledge Distillation

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***Abstract*— Scene classification is a fundamental task for numeral remote sensing (RS) applications, which aims to assign semantic labels to image patches. Although deep neural networks (DNNs) demonstrated unique strength in scene classification, their performances are still limited due to the lack of training samples in the RS field. Recent studies show that the performance of scene classification can be improved by taking advantage of the knowledge transferred from models pretrained on RGB images. However, the modalities' differences between input images hinder the knowledge transfer across models, especially when the input of the models has distinct spectral bands. To tackle the challenges, we propose a cross-modal knowledge distillation framework to improve the performance of multispectral scene classification by transferring the prior knowledge from teacher models pretrained on RGB images to the student network with limited samples. Moreover, a teacher assistant (TA) network is introduced to further improve the classification performance by bridging the gap between the teacher and student networks. The proposed strategy is evaluated on models with multimodality inputs with distinct spectral bands and demonstrates superior performance compared to the state-of-the-art methods.**

***Index Terms*— Knowledge distillation (KD), multispectral imagery (MSI), scene classification.**

I. Introduction

SCENE classification is one of the fundamental problems in remote sensing (RS) imagery analysis, which aims to label scene patches with one or more semantic categories. It contributes significantly to a wide range of applications, including land-cover/land-use investigating [1]–[8] and agriculture [9]–[11]. Nowadays, satellites can provide multispectral imageries (MSIs) with rich spectral characteristics, which has the potential to distinguish similar scenes better than traditional RGB images. Thus, MSI-based scene classification has attracted increasing attention recently.

Traditional scene classification studies mainly use shallow handcrafted features or mid-level representations to distinguish the types of scenes, which usually have limited performance in complex scenarios. Recently, a deep neural network (DNN) has revolutionized the field of computer vision and demonstrated unique strength in RS scene classification tasks [12]–[14]. However, well-performing DNN models are mostly built with high-capacity convolutional neural networks (CNNs) developed for computer vision tasks. These CNNs are usually constructed with a tremendous amount of weight parameters. For example, the feature extraction layers of the VGG16 (CNN network) [15] adopted in [16] have nearly 14.7M parameters. To optimize the scene classification performance in the RS field, the DNN models have to be trained with an adequate amount of labeled samples to prevent overfitting. Nonetheless, there is no *ImageNet*-like [17] large-scale dataset available in the RS field, and the data annotation in this field is resource-consuming and requires certain expert knowledge. Thus, in most scenarios, labeled samples are usually inadequate, which may lead to the performance degradation of the DNN-based classification models.

To address the challenge above, a common strategy [12], [18], [19] is to adopt the classification models pretrained on an existing computer vision dataset (e.g., *ImageNet*) and fine-tunes them on the RS datasets with limited labels. However, the profound differences between computer vision images and RS imageries may limit the performance of fine-tuning. Most pretrained classification models are designed for input images with three channels, i.e., R, G, and B. Conversely, RS imagery obtained by sensors on a variety of platforms often holds multiple spectral bands. To feed imageries with multiple spectral bands to a pretrained model with three input channels, we have to abandon information in certain spectral bands, which may waste potential information that can distinguish similar scenes.

This, naturally, raises a question: can we take advantage of the prior knowledge learned from RGB images to classify scenes with MSIs?

Recent advances in knowledge distillation (KD) methods in the computer vision field have the potential to address this problem. The idea of KD was to compress large models into smaller ones without losing much performance [20], which is introduced by Hinton *et al.* [21] in 2015. Recent KD methods include basic teacher–student learning [21], mutual learning [22], KD with teacher assistant (TA) [23], and self-training [24], which are characterized by the definitions of knowledge, distillation schemes, and frameworks. These KD methods mainly focus on model compression, where a large pretrained network (teacher) is used to train a lightweight network (student), and the input images of both models belong

Manuscript received January 17, 2022; revised April 21, 2022; accepted May 7, 2022. Date of publication May 11, 2022; date of current version May 23, 2022. This work was supported by the National Natural Science Foundation of China under Grant 41925006. *(Corresponding author: Liqiang Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2022.3174352

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to the same modality with identical channels. Nonetheless, the possibility of KD between different modalities with nonidentical channels has not yet been fully explored, such as in the scenarios where two scene classification models use different spectral bands of the same scene patches as the input.

Beyond model compression, we believe that KD has the potential to improve the scene classification performance by transferring the knowledge between multimodalities. As analyzed before, transferring the prior knowledge directly between two models with different numbers of input channels (e.g., RGB, MSI) is not viable due to the differences in the model structures. However, through KD, the multispectral student network can learn the prior knowledge from the teacher network pretrained on RGB images by mimicking its prediction, regardless of the differences between network structures. This strategy, thus, helps us break the input restrictions brought by pretrained networks while allowing the use of MSI as input, thus improving the performance of an MSI-based scene classification model.

In this article, we propose a cross-modal teacher–student KD strategy to improve the training of multispectral scene classification models by teacher networks pretrained and fine-tuned on RGB images. Moreover, we show that a large performance gap between the teacher and student networks will lead to the performance degradation of the student network, which can be alleviated by multistep KD. The contributions of the proposed method can be summarized as follows.

1) First, a cross-modal KD method is proposed to improve the performance of MSI-based scene classification through the teacher models pretrained on RGB images.
2) Second, to bridge the gap between student and teacher networks, an intermediate-sized TA network is introduced to perform two-step KD, which further improves the performance of the MSI-based scene classification.
3) Third, to tackle the problem of performance degradation when the input of the student and teacher models have nonoverlapping bands, the KD between two models with totally different spectral bands as input is also explored and demonstrates that the proposed cross-modal KD is still effective in such scenarios.

The remainder of this article is organized as follows. Section II provides a brief review of related works. Section III introduces the proposed KD framework. Section IV presents experimental results on two publicly available multispectral scene imagery datasets. Finally, Section V concludes this article.

II. Related Work

A. Remote Sensing Scene Classification

The objective of RS scene classification is to correctly assign semantic labels to the images based on their contents. There have been numerous traditional methods developed using low-level visual features (e.g., [25]–[27]), as well as mid-level visual representations (e.g., [3] and [28]–[31]) to address the problem of RS scene classification. Recent studies of scene classification using DNN models have achieved better performance than the aforementioned methods. For example, Penatti *et al.* [32] and Cheng *et al.* [33] investigated strategies using pretrained deep CNNs for RS scene classification tasks. Zhang *et al.* [34] performed classification by combining multiple DNNs, while Chaib *et al.* [35] proposed a feature fusion strategy to integrate features from different layers of CNN for very high resolution (VHR) scene classification. Cheng *et al.* [36] used a CNN with metric learning regularization term, and Yu *et al.* [37] proposed a feature learning framework for aerial scene classification by combining generative adversarial networks (GANs) with an attention mechanism. Teng *et al.* [16] introduced an adversarial domain adaptation method that trained the scene classifier in a semisupervised manner. Lu *et al.* [38] introduced a multisource compensation network to form the categories of the target domain with multiple complementary source domains. For small-sample-size RS scene classification tasks, Mei *et al.* [39] proposed a sparse representation-based framework with deep feature fusion, using well-trained DNNs for feature extraction.

B. Knowledge Distillation

1) Definition of Knowledge: Existing KD frameworks can be categorized into four types based on the definition of knowledge. The first type of knowledge is defined as the logits of a DNN model. For example, Hinton *et al.* [21] proposed using the logits (network outputs before softmax layer) as knowledge in an image classification task. By measuring the temperature-T-softened similarity between the logits of teacher and student networks, a student network can distill knowledge from a teacher network by adopting its predicted logits as a soft target [40]. These soft targets generated by the inference process of the teacher tend to be easier for students to learn compared to the hard targets, i.e., the "yes or no" form of target values given by the ground truth, in vanilla supervised learning scenario.

The second type of knowledge is drawn from intermediate representations of a network. As first introduced in [41], feature-based KD methods [42]–[46] aim to match the final output of the teacher/student network and the feature activations of certain intermediate layers.

The third type of knowledge comes from the relationship between layers of a network. A number of such definitions of knowledge have been proposed using, for example, the Gram matrix [47], singular value decomposition [48], multihead attention mechanism [49], and information flow between hint layer pairs of a network [50].

Finally, knowledge can also be defined by the relationship between samples. For example, Passalis and Tefas [51] proposed a novel method that minimizes the divergence between the probability distributions of the representations extracted from the teacher network and the student network. Park *et al.* [52] proposed to preserve the feature similarities of samples in the intermediate layers of the teacher networks while training student networks. Tung and Mori [53] proposed to distill and transfer the pairwise knowledge from the teacher's activation.

2) Distillation Scheme: Managing the flow of knowledge transfer in students' training process is an essential technique

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in KD methods. Because the teacher network is the source of knowledge, we summarize existing distillation schemes based on their arrangement for the teacher network.

For off-line distillation schemes (e.g., [21]), teacher networks are pretrained with a training set before the KD process initiates. When training student networks, teacher networks only infer and provide logits or other hint information for the students; no further parameter optimization for the teacher network is required.

In some scenarios, it is difficult to obtain a teacher network that performs well. Online distillation schemes can be useful in such cases, where both teacher and student networks are updated simultaneously. For example, Zhang *et al.* [22] introduced a mutual KD framework in which two networks can exchange the roles of teacher and student during training. In [54], multiple models are trained in parallel, while anyone model can distill knowledge from other ones.

3) Teacher–Student Framework: Designing the relationship between teacher and student networks is also an important part of the KD approach. KD from one teacher to one student is the most commonly used setting. The problem with such a setting is that, when there is a major gap in performance between student and teacher networks, the student can hardly benefit from KD practice. Several studies have provided possible solutions to this issue. The mutual learning scheme proposed in [22] involves two student networks with similar performance, while learning from each other can benefit both networks. In [23], the use of TA networks turns a one-step KD into a multistep KD, closing the gap in performance between teacher and student networks while improving the final performance of the student.

III. Methodology

A. Problem Formulation

Scene classification aims to assign a set of appropriate labels to scene patch samples. Assuming that we have a multispectral scene imagery dataset with m scene patches, $X = \{X_1, X_2, \ldots, X_m\}$, then there exist k scene categories in this dataset, and each scene patch has label $Y = \{y_1, y_2, \ldots, y_k\}$. If this scene patch belongs to the jth category, then $y_j = 1$; otherwise, $y_j = 0$. For multiclass single-label scene classification scenarios, each scene patch can have only one label: $\sum_{j=1}^{k} y_j = 1$, while, for multiclass multilabel scene classification scenarios, each scene patch can have one or more labels: $\sum_{j=1}^{k} y_j \geq 1$.

B. One-Step Cross-Modal KD

First, we consider an off-line KD strategy. Before the KD process initiates, a pretrained RGB image classification network is fine-tuned on the target multispectral RS imagery dataset. During fine-tuning, the best-performing model is chosen through validation and testing, and its parameters are preserved. In the later KD process, a model with these parameters is regarded as a teacher network $\mathcal{T}$.

Based on specific imagery application tasks, we design the proposed scene classification networks with the required input patch size and bands. We consider this model to be student network $\mathcal{S}$. Because this student network might not have a typical input size $(224, 244)$ or RGB band input, pretrained parameters may be unavailable for this network. Therefore, training on task datasets determines the final performance of student network $\mathcal{S}$, and we intend to improve it with the proposed KD framework.

Considering that pretrained teacher networks and task-oriented student networks might not share similar structures, in this study, we do not define knowledge in the KD process computed from intermediate features as in [41]–[43]. Instead, we choose the logit output [21] of the teacher network as the knowledge to transfer; thus, we can avoid the difference between the network structures of teachers/students, and the only concern left is the performance gap between teachers/students in task scene classification scenarios.

Fig. 1 shows the proposed one-step KD framework. While training the student network, two loss functions are used. The first loss comes from the supervision information of the scene imagery dataset. Based on ground-truth labels Y_{GT} from the task dataset, a supervised learning loss can be computed with the label prediction $\mathcal{S}(X)$ of the student network

$$L_{\mathrm{SL}} = \mathcal{H}(Y_{\mathrm{GT}}, \mathcal{S}(X)). \tag{1}$$

For the loss criterion $\mathcal{H}$, cross-entropy (CE) is chosen for multiclass single-label classification scenarios, while, for multiclass multilabel classification scenarios, we use binary CE (BCE). The second loss $L_{\mathcal{TS}}$ originates from the KD process, where the student network attempts to mimic the output of the teacher network and is computed in the same manner as the first loss

$$L_{\mathcal{TS}} = \mathcal{H}(\mathcal{S}_T(X), \mathcal{T}_T(X)) \tag{2}$$

where $\mathcal{S}_T(X)$ and $\mathcal{T}_T(X)$ are the logits from student and teacher networks, respectively, and softened by temperature parameter T [21]. This softening procedure aims to reduce the difference between two network predictions, thus reducing the difficulty in students' KD learning process. Finally, training the student network combines supervised learning and the KD process, and is performed by minimizing the following loss function:

$$L_{\mathrm{student}} = (1 - \lambda) * L_{\mathrm{SL}} + \lambda * L_{\mathcal{TS}} \tag{3}$$

where λ controls the proportion of supervised learning and KD process in effecting the student network's training. When $\lambda = 0$, this training paradigm becomes common supervised learning. When $\lambda = 1$, the ground-truth information is ignored, and the student network is trained by completely relying on mimicking the prediction of the teacher network. Such a training strategy is called *matching logits* [21] and can be used in semisupervised learning scenarios.

Because RS imagery can have data from multiple spectral bands, in this study, we do not require the teacher and student network to share the same group of bands as input, which is a setting different from previous research [22], [23], [55]. The proposed KD framework operates effectively if the input imagery data of two networks are directed to the same scene patch.

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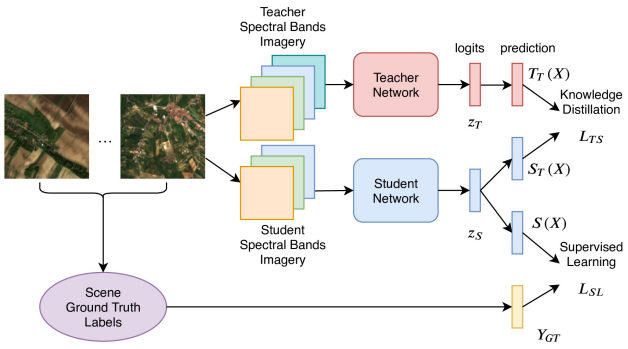


Fig. 1. Proposed one-step cross-modal KD framework.

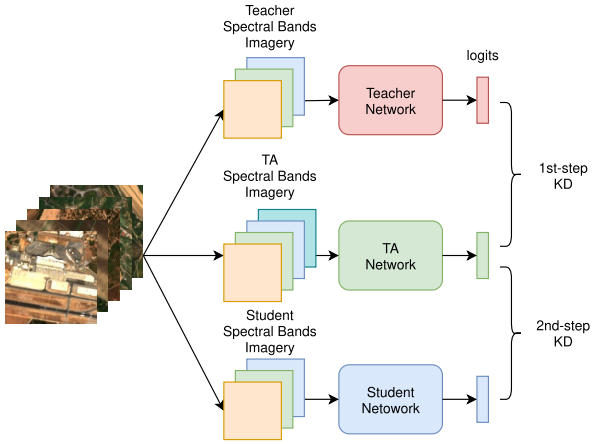


Fig. 2. Proposed two-step KD framework using one TA network.



Fig. 3. Example images of 43 scene classes in the BigEarthNet dataset.

C. Two-Step Cross-Modal KD With Teacher Assistant

In some scene classification scenarios, the performance of teacher networks with pretrained parameters is considered superior compared to that of a task-oriented student network with no pretrained parameters available. In addition, the structure of the student network might not have the capacity to mimic the prediction of the teacher. The result of the KD process can be compromised when distilling from the prediction of a teacher network that is too near the ground truth.

We find inspiration in the multistep teacher–student KD strategy proposed in [23] for this performance gap issue. By introducing a TA network into the KD framework, the student network can achieve better performance when learning to TA compared to distilling directly from the teacher network. For multispectral scene imagery classification scenarios in this study, we can design a series of classification models with different groups of bands as input. An intermediate scene classification model with additional bands as input can serve as a TA to draw close the gap between the RGB pretrained teacher model and the task-required student model.

The proposed two-step KD strategy is formulated as follows. For a scene classification task using MSI, it is possible to formulate a series of classification network models with different groups of bands as input. Through supervised learning and testing, the best-performing model among them is chosen as TA. The first step KD is conducted between the pretrained RGB teacher and the chosen TA network, and then, the second step KD is conducted between the TA and the task-required student network model. More TA networks can be added to this strategy. For each step of KD, the training loss L is formulated in a similar manner as the abovementioned L_{student}. Note that L_{TS} is defined between logits from teacher/TA or TA/student networks, depending on the specific step of KD. Fig. 2 shows a KD scenario using one TA network.

IV. Experiments

A. Dataset Description

To evaluate our proposed method, we perform experiments on two benchmark datasets: BigEarthNet [56] and NaSC-TG2 [19]. They are designed for RS image scene classification and provided MSI data for each scene patch.

The BigEarthNet dataset[1] [56] is a *multiclass multilabel* scene classification benchmark and consists of 590 326 multispectral RS imagery patches obtained from the Sentinel-2 platform. Each patch is annotated by multiple labels from 43 land-cover categories. Examples of all 43 land-cover categories in the dataset are shown in Fig. 3. From all 13 bands of data provided for each scene patch, we choose the RGB and NIR spectral band images for experiments since they have the same spatial resolution (10 m), and for each band, the size of the image patch is 120 × 120. Following the instructions in [56], we remove 70 987 images that are fully covered by seasonal snow, clouds, and cloud shadows in the following experiments.

The NaSC-TG2 dataset[2] is a *multiclass single-label* benchmark for RS natural scene classification from Tiangong-2 remotely sensed imagery that contains 20 000 scene patches, which are equally divided into ten natural scene classes. Examples of the included scene classes are shown in Fig. 4. Each scene patch has a size of 128 × 128, and its spatial resolution is 100 m. In addition to RGB images, corresponding scene images of 14 programmable visible and near-infrared (VNIR) channels [57] are also provided. As introduced in [58], the detailed information of these 14 channels is shown in Table I.

[1] http://bigearth.net/
[2] http://www.msadc.cn/main/setsubDetail?id=1370312964720037889

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TABLE I
CENTER WAVELENGTH AND BANDWIDTH INFORMATION OF 14 VISIBLE LIGHT AND NEAR INFRARED CHANNELS IN THE NASC-TG2 DATASET

Channel	Center Wavelength (nm)	Bandwidth (nm)
V1	413	403-423
V2	443	433-453
V3	490	480-500
V4	520	510-530
V5	565	555-575
V6	620	610-630
V7	665	655-675
V8	682.5	677.5-687.5
V9	750	740-760
V10	820	810-830
V11	865	845-885
V12	905	895-915
V13	940	930-950
V14	980	970-990

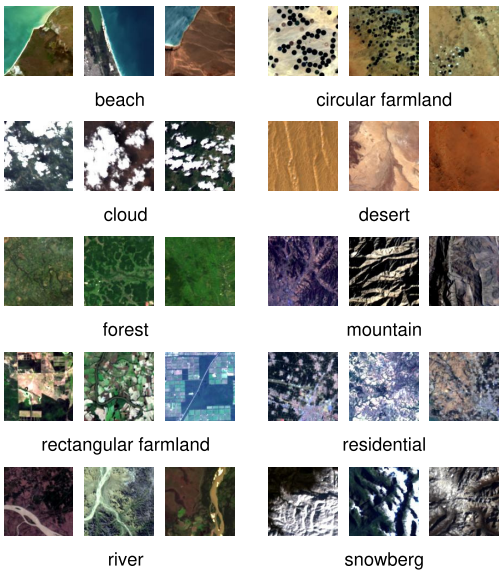


Fig. 4. Example images of ten scene classes in the NaSC-TG2 dataset.

B. Experimental Design and Evaluation Metrics

To evaluate the proposed cross-modal KD framework in one- and two-step settings, we use all 43 scene classes in the BigEarthNet dataset, with all valid scene patches after the removal of seasonal snow, clouds, and cloud shadows. By dividing the dataset into training, validation, and testing sets with 60%, 20%, and 20% scene patches, respectively, we created a 43-class multilabel scene classification scenario. For quantitative comparison, the scene classification performance is evaluated with three widely used metrics, i.e., precision (P), recall (R), F1, and F2. These metrics are defined as follows:

$$P = \frac{\text{tp}}{\text{tp} + \text{fp}}$$
$$R = \frac{\text{tp}}{\text{tp} + \text{fn}} \tag{4}$$
$$F^{\beta} = \frac{\left(1 + \beta^2\right) P * R}{\beta^2 * P + R}$$

where F1 is set as the primary evaluation metric to balance precision and recall for experiments on the BigEarthNet dataset. In each experimental setting, the set of parameters with the highest F1 score in validation is preserved during iterative training.

Further experiments are conducted on the NaSC-TG2 dataset to evaluate the effectiveness of the proposed KD framework in scenarios where teacher and student networks do not share any input band at all. For each natural scene class in the NaSC-TG2 dataset, 60%, 20%, and 20% of the scene patches are randomly selected as the training set, the validation set, and the testing set, respectively. For evaluation, we choose overall accuracy (OA) as a metric, defined as follows:

$$\text{OA} = \frac{N_{\text{true}}}{N} \tag{5}$$

where N_{true} is the number of correctly predicted scene patches and N is the total number of scene patches participating in the test set. In each experimental setting, the set of parameters with the highest OA in validation is preserved during iterative training.

We evaluate the proposed method from three perspectives. First, we conduct experiments on the BigEarthNet dataset to evaluate the effectiveness of the proposed one- and two-step cross-modal KD strategies. Second, to explore KD between models using different spectral band imagery, we further conduct experiments on the NaSC-TG2 dataset, using the VNIR spectral band imageries with the same spatial resolution. In addition, we compare the performance of our proposed one- and two-step KD methods with other state-of-the-art deep learning-based methods on the NaSC-TG2 dataset.

C. Classification Results With One-Step Cross-Modal Knowledge Distillation

The one-step teacher-to-student cross-modal KD is the core of the proposed framework. To evaluate its effectiveness in the MSI field, we first conduct scene classification experiments on the BigEarthNet dataset. The feature extraction layers of two pretrained networks, i.e., a large pretrained network MobileNetV3-Large and an intermediate-sized network MobileNetV3-Small introduced in [59], are employed as the backbones to evaluate the proposed cross-modal KD scheme. The structures of the backbones of MobileNetV3-Small and MobileNetV3-Large networks are presented in Tables II and III, respectively. The teacher models are constructed with the feature learning layers of the MobileNetV3-Large and MobileNetV3-Small models and concatenated with fully connected (fc) layers and sigmoid activation functions.

Each of these two teacher networks is fine-tuned five times on the training samples of the BigEarthNet with RGB images only, and the parameters of the best performing model in the testing stage are retained for the following KD experiments. BCE is chosen as the criterion metric $\mathcal{H}$ for supervised learning and KD.

While fine-tuning the teacher networks, the scene patch size is resized to (224, 224); the batch size is set to 32. The network is optimized by the Adam optimizer, while the learning

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TABLE II

STRUCTURE OF THE MOBILENETV3-SMALL FEATURE BACKBONE NETWORK

operator	exp size	SE block	nonlinearity	stride
conv2d, 3*3	-	-	h-swish	2
bneck, 3*3	16	yes	ReLU	2
bneck, 3*3	72	-	ReLU	2
bneck, 3*3	88	-	ReLU	1
bneck, 5*5	96	yes	h-swish	2
bneck, 5*5	240	yes	h-swish	1
bneck, 5*5	240	yes	h-swish	1
bneck, 5*5	120	yes	h-swish	1
bneck, 5*5	144	yes	h-swish	1
bneck, 5*5	288	yes	h-swish	2
bneck, 5*5	576	yes	h-swish	1
bneck, 5*5	576	yes	h-swish	1
conv2d, 1*1	-	yes	h-swish	1

TABLE III

STRUCTURE OF THE MOBILENETV3-LARGE FEATURE BACKBONE NETWORK

operator	exp size	SE block	nonlinearity	stride
conv2d	-	-	h-swish	2
bneck, 3*3	16	-	ReLU	1
bneck, 3*3	64	-	ReLU	2
bneck, 3*3	72	-	ReLU	1
bneck, 5*5	72	yes	ReLU	2
bneck, 5*5	120	yes	ReLU	1
bneck, 5*5	120	yes	ReLU	1
bneck, 3*3	240	-	h-swish	2
bneck, 3*3	200	-	h-swish	1
bneck, 3*3	184	-	h-swish	1
bneck, 3*3	184	-	h-swish	1
bneck, 3*3	480	yes	h-swish	1
bneck, 3*3	672	yes	h-swish	1
bneck, 5*5	672	yes	h-swish	2
bneck, 5*5	960	yes	h-swish	1
bneck, 5*5	960	yes	h-swish	1
conv2d, 1*1	-	-	h-swish	1

TABLE IV

CLASSIFICATION PERFORMANCE OF TEACHER NETWORKS RETRAINED ON THE BIGEARTHNET DATASET

Teacher Network	P(%)	R(%)	F1	F2
MobileNetV3-Small	84.59	79.66	0.8205	0.8060
MobileNetV3-Large	85.07	82.35	0.8369	0.8288

rate is fixed to 0.0001. The training procedure continues for 50 epochs, and we repeat the experiment five times. For each teacher network, the results of the best performing model are shown in Table IV.

For the student network, we modified the ResNet18 model [60] and created a smaller ResNet with eight convolutional layers. In this study, we refer to this model as ResNet8. The number of input neurons of the ResNet8 model is adjusted according to the number of bands of the input images in the BigEarthNet dataset. Similar to the teacher networks, we discarded the original classifier of the ResNet model and concatenated an fc with a sigmoid activation function to the end of the feature learning layers. To evaluate the effectiveness of the cross-modal transfer, both student networks with three-channel input (RGB) and four-channel input (RGB/NIR) are tested in the following experiments. The network structure of ResNet8 is demonstrated in Table V.

TABLE V

MODIFIED RESNET8 NETWORK STRUCTURE

Layer	Attributes
(input)	size: (120,120), channels: 3 or 4
Conv2d	filters: 64, kernel size: (5,5), stride: 2
bn	
relu	
maxpool	kernel size:(3, 3), stride: 2
Residual Block * 1	planes: 128, stride: 2
Residual Block * 1	planes: 128, stride: 2
Residual Block * 1	planes: 128, stride: 2
AvgPool2d	kernel size: (4, 4)
fc1	in: 512, out: 128
fc2	in: 128, out: 43
sigmoid	

TABLE VI

CLASSIFICATION PERFORMANCE OF STUDENT NETWORKS TRAINED IN SUPERVISED LEARNING MANNER WITH GROUND TRUTH OF THE BIGEARTHNET DATASET

Student Network	P(%)	R(%)	F1	F2
ResNet8-RGB	82.58	74.56	0.7835	0.7603
ResNet8-RGBNIR	83.21	76.59	0.7975	0.7786

We first evaluate the two student networks in a fully supervised manner without KD. In this experiment, the batch size is set to 32; optimization is conducted with the Adam optimizer, while the learning rate is fixed to 0.0001. The networks are trained for 50 epochs, and the experiment is repeated five times. The averaged performance is reported in Table VI. These reported performances are regarded as baselines, which will be improved with KD.

1) Effects of the One-Step KD: With the teacher and student networks, we conduct experiments with one-step KD. For each pair of teachers/students, the KD parameter λ is set to 0.5, 0.9 and 1.0; the temperature parameter T is set to 1.0 and 10.0. Note that, when λ is set to 1.0, the student network will be updated based on the logits transferred from the teacher network only, while, when $\lambda = 0$, the network will be trained in a supervised minor (baseline). The results of the one-step KD are shown in Table VII, where the best results in terms of F1 for each student network are marked in red. Based on the experimental results, we draw two conclusions: 1) the proposed cross-modal distillation learning strategy performs better than supervised learning and 2) the distillation is more effective when the performance gap between the teacher and student is small.

By comparing the performance of the listed KD settings with the supervised learning baseline, we observe marked improvements. For the ResNet8-RGB student network, the best performance is achieved with $\lambda = 0.9$ and $T = 10.0$, where the proposed KD framework boosted the F1 score over the baseline by 0.009. For the ResNet8-RGBNIR student network, setting $\lambda = 0.5$ and $T = 1.0$ or $\lambda = 0.9$ and $T = 10.0$ achieved the best F1 score, which was 0.023 above the corresponding baseline. Thus, we conclude that the proposed one-step KD framework yields marked improvements in this multispectral scene classification scenario.

It is worth noting that, although the MobileNetV3-Large teacher network achieves better performance (F1 = 0.8369)

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TABLE VII
RESULTS OF ONE-STEP KD ON THE BIGEARTHNET DATASET

Student Network & Bands	Teacher Network & Bands	λ	T	P(%)	R(%)	F1	F2
ResNet8-RGB		0		82.58	74.56	0.7835	0.7603
	MobileNetV3-Small-RGB	0.5	1.0	83.40	75.40	0.7920	0.7688
			10.0	83.37	74.21	0.7852	0.7588
		0.9	1.0	84.10	74.86	0.7921	0.7665
			10.0	82.75	76.12	0.7925	0.7735
		1.0	1.0	83.49	75.02	0.7913	0.7673
			10.0	82.79	75.66	0.7907	0.7698
	MobileNetV3-Large-RGB	0.5	1.0	83.44	74.83	0.7890	0.7640
			10.0	82.24	75.17	0.7855	0.7649
		0.9	1.0	83.12	74.35	0.7849	0.7595
			10.0	82.83	74.91	0.7867	0.7637
		1.0	1.0	83.70	73.96	0.7853	0.7573
			10.0	80.96	77.03	0.7894	0.7778
ResNet8-RGBNIR		0		83.21	76.59	0.7975	0.7786
	MobileNetV3-Small-RGB	0.5	1.0	84.87	76.63	0.8054	0.7815
			10.0	82.92	76.97	0.7983	0.7809
		0.9	1.0	84.14	76.55	0.8017	0.7796
			10.0	82.70	78.48	0.8054	0.7929
		1.0	1.0	84.73	75.69	0.7995	0.7734
			10.0	83.60	76.68	0.7999	0.7797
	MobileNetV3-Large-RGB	0.5	1.0	83.81	76.16	0.7980	0.7757
			10.0	84.11	76.03	0.7986	0.7752
		0.9	1.0	85.35	75.00	0.7984	0.7686
			10.0	83.45	76.68	0.7992	0.7795
		1.0	1.0	83.96	75.71	0.7962	0.7723
			10.0	82.83	77.02	0.7982	0.7812

compared to that achieved by MobileNetV3-Small (F1 = 0.8205), it does not mean that it is a better teacher. As shown in Fig. 5, MobileNetV3-Small, whose performance is closer to that of two students, is more effective in KD under the same parameter settings. These results confirm the findings in [23], and a better performing model does not guarantee a more effective teacher; when the gap between teacher and student is greater than a threshold, the effectiveness of the KD drops.

To understand the influence of the distillation process when training the student network, experiments with $\lambda = 1$ were conducted to exclude the influence of supervised learning, as shown in Table VII. When MobileNetV3-Small serves as the teacher network, the student network ResNet8-RGB did not achieve the best performance. However, it still achieves a better F1 score compared to that achieved with supervised learning. However, when a better-performing MobileNetV3-Large network acts as a teacher, the performance gap between the teacher and the student increases, while the F1 scores achieved by both student networks are lower than that achieved with the MobileNetV3-Small network.

Based on these observations, we infer that, for scene classification scenarios, the KD objective is an acceptable alternative to supervised learning. With proper teacher selection, a scene classification network trained with a pure KD setting can achieve performance on par with or even superior to that achieved by supervised learning. When combining the KD objective with supervised learning in an appropriate proportion, the proposed framework can achieve even better results.

2) Parameter Analysis: During KD training, the temperature T is used to control the difference measurement between predictions. To evaluate the effect of temperature T in the proposed scene classification framework, we first fix the parameter λ and adjust T. When $\lambda = 1.0$, a higher temperature T leads to a better performance in most cases

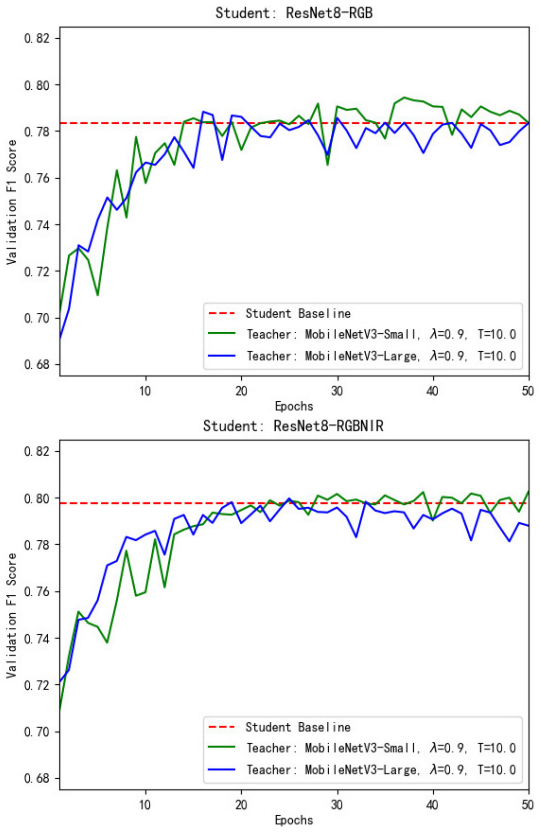


Fig. 5. Validation F1 score curves in KD with two teacher networks.

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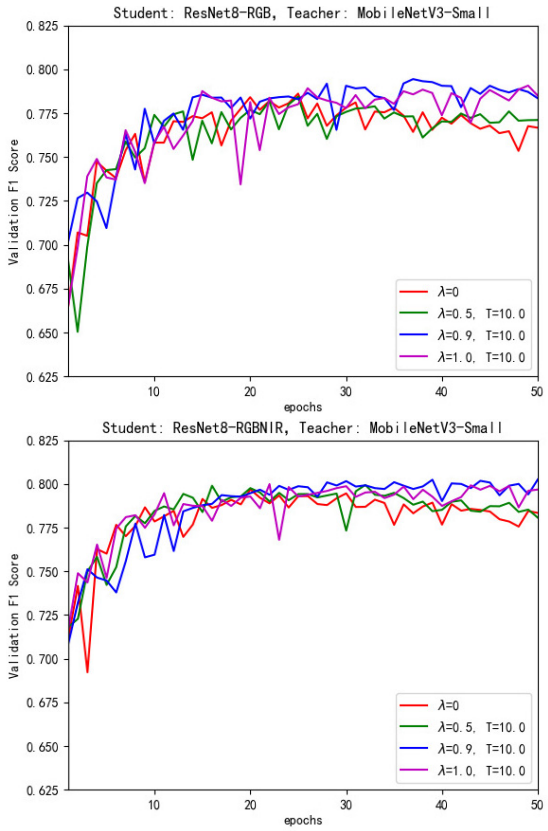


Fig. 6. Validation F1 score curves in KD with different λ settings.

except when ResNet8-RGB is selected as the student, and MobileNetV3-Small is selected as the teacher. If supervised learning and KD are equally important (i.e., $\lambda = 0.5$), a high temperature does not improve the training. With $\lambda = 0.9$, the KD contributes more to the optimization; both students achieve the best performance.

As shown in Table VII, the selection of temperature T has a strong impact on the performance of the proposed KD framework. T should be determined based on the performance gap between teacher/student, as well as the student's degree of acceptability with respect to the teacher's prediction output.

The effects of λ are closely related to the effectiveness of both supervised learning and distillation processes in specific scene classification scenarios. The results of the proposed KD framework with different λ settings are shown in Fig. 6. Compared to supervised learning, optimizing the networks with KD achieves better performance. It is, thus, reasonable that the best performance is achieved with KD having the greater weights ($\lambda = 0.9$) for these two teacher/student pairs.

When selecting λ, we suggest that it should be determined by the reliability of supervised learning on a given training set. In scenarios where scene imagery samples are short and supervised learning cannot yield desirable results, setting λ closer to 1.0 has better potential if an acceptable teacher network is available.

The model size and average time consumption in the inference stage for different network models involved in the KD experiments are shown in Table VIII. The proposed

TABLE VIII
MODEL SIZE AND TIME CONSUMPTION IN THE INFERENCE STAGE FOR NETWORKS IN THE EXPERIMENTS WITH THE BIGEARTHNET DATASET

Network & Bands	params	Inference Time (ms)
MobileNet-V3-small-RGB	951.8K	5.577
MobileNet-V3-Large-RGB	3.013M	6.962
ResNet8-RGB	930.4K	1.462
ResNet8-RGBNIR	932.0K	1.475

cross-modal KD scheme is able to transfer the knowledge from computational expensive models (MobileNet-V3-Large-RGB/MobileNet-V3-small-RGB) to computational efficient neural networks (ResNet8-RGB/ResNet8-RGBNIR), which saves around 20% of the inference time cost by the teachers, with improved accuracy.

D. Classification Results With Two-Step KD

In addition to one-step KD, we further improve scene classification models with multistep KD when there are no overlapping bands between the input of the teacher and student model. Experimental results in one-step KD on the BigEarthNet dataset suggest that the performance of the student networks is poor if the gap between teacher/student performance is large, as discussed in [23]. There is room for intermediate TA networks to further improve the performance of the students. In addition, the RGB-NIR four-band student model achieves better performance than the RGB three-band model regardless of the parameters. Thus, this model can be employed as a TA that fills the performance gap between the pretrained model and students.

Thus, we conduct experiments using these teacher/student network models to evaluate the performance of the two-step KD in the proposed method. We employ the MobileNetV3-Large and MobileNetV3-Small as the teacher networks with RGB input and the four-band ResNet8-RGBNIR network as a teaching assistant network to facilitate the training of the student network (three-band ResNet8-RGB). The results are shown in Table IX, where the best result in terms of the student's F1 metric is marked in red.

Compared to the best performance (F1 = 0.7925) achieved by the ResNet8-RGB student model in one-step KD, the two-step KD further improves the performance of the scene classification. As shown in Fig. 7, the F1 curves of the two-step KD from both teachers are superior to the ResNet8-RGB baseline, which indicates that the proposed two-step KD strategy with a multispectral TA network is effective.

E. Cross-Modal Knowledge Distillation Using Multispectral Imagery With Different Spectral Band

In previous experiments, although the input modalities of the teacher and student models are different, they still share the same RGB bands. However, in real applications, it is normal that the input of the two modalities covers very different wavelength ranges, e.g., RGB and VNIR. To further evaluate the effectiveness of the proposed cross-modal KD, we conduct

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TABLE IX
RESULTS OF TWO-STEP KD ON THE BIGEARTHNET DATASET

Path No.	Teacher Network	1st-step KD parameters	TA Network	2nd-step KD parameters	Student Network
1	MobileNetV3-Large F1 = 0.8369	λ = 0.9 T = 10.0	ResNet8-RGBNIR F1 = 0.7992	λ = 0.9 T = 10.0	ResNet8-RGB F1 = 0.7950
2	MobileNetV3-Small F1 = 0.8205	λ = 0.9 T = 10.0	ResNet8-RGBNIR F1 = 0.8054	λ = 0.9 T = 10.0	ResNet8-RGB F1 = 0.7949

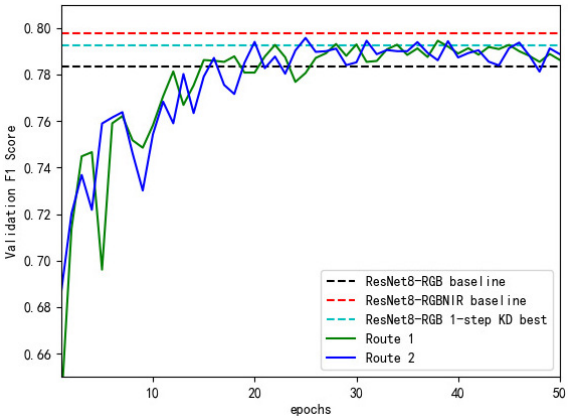


Fig. 7. Validation F1 score of two-step KD.

experiments on the NaSC-TG2 dataset with both RGB and VNIR images.

In this study, two groups of experiments are conducted to evaluate the effectiveness of knowledge transfer. In the first group, three bands (2, 5, 8) of the VNIR images are selected as the input of the student image, while, in the second group, six bands (2, 3, 4, 5, 8, 9) of the VNIR images are selected as the input of the student network. In both groups, the inputs of the teacher network are RGB images.

To avoid the bias caused by the capacity differences between the teacher and student models during the KD procedure, both the teacher and student models use the MobileNetV3-Small as the prototype network. The number of the input neurons in the student networks is adjusted based on the number of the input bands. In this study, the student model is named MobileNetV3-alter-Small. The teacher/student networks are concatenated with an fc layer, as well as a softmax layer to accommodate the ten-class single-class scene classification scenario, while CE is used as a criterion for both the supervised learning loss L_{SL} and the KD loss $L_{\mathcal{TS}}$. For the experimental settings, the batch size is set to 40, and optimization is conducted with the Adam optimizer, while the learning rate is initialized to 0.0004 and decays to half for every 20 epochs. For each setting, the training procedure goes for 100 epochs, and we repeat the experiment five times.

First, we test the performance of the teacher and student networks before KD. As reported in Table X, the teacher network MobileNetV3-Small pretrained and fine-tuned on the RGB images achieved leading performance with an OA of 0.9908. For the models using VNIR channels as input, under the same supervised training settings, the chosen six-VNIR-band model managed to score higher than the three-VNIR-band model, which indicates that the six-VNIR-band imagery can provide richer information than the three-VNIR-band images. However, there is a gap between the performance of the student models using VNIR input imagery and the teacher model pretrained on RGB images, and we intend to determine whether the knowledge can be distilled between these models using distinct imagery inputs.

TABLE X
RESULTS OF VANILLA SUPERVISED TRAINING FOR TEACHER/STUDENT MODELS ON THE NASC-TG2 DATASET

Network	Input Bands	Overall Accuracy (OA)	
		Average	std
MobileNetV3-Small pretrained	RGB	0.9908	0.0014
MobileNetV3-alter-Small	VNIR channels V2,5,8	0.9078	0.0146
MobileNetV3-alter-Small	VNIR channels V2,3,4,5,8,9	0.9275	0.0041

1) Effects of the One-Step KD: The MobileNetV3-Small model pretrained on RGB images with OA = 0.9925 is employed as the teacher network to improve the performance of the two student networks with VNIR input imagery. Temperature T is set to 1.0, 10.0, and 50.0 to explore the effect of softening prediction. λ is set to 0.5, 0.9, and 1.0. Results in terms of average OA and its standard deviation are shown in Table XI. The best results achieved for each student network are marked in red.

As shown in Table XI, the proposed cross-modal KD framework is still effective in this KD-across-bands' scenario. As for both students, the result of KD in a variety of settings surpasses the supervised learning baseline. For the three-VNIR-band student, the best performance is achieved with $\lambda = 1.0$ and $T = 10.0$, where the supervised learning loss with ground-truth labels is omitted. Conversely, although, in most settings, the six-VNIR-band student did not perform better than the baseline, with $\lambda = 0.9$ and $T = 10.0$, it still surpassed the baseline. Based on the observations, we conclude that, if the input scene imagery is informative (e.g., six-band VNIR), the supervised learning loss should be given higher weight in the proposed KD framework; otherwise (e.g., three-band VNIR), the distillation procedure should be weighted more. For both students, $T = 10.0$ leads to a better performance in most cases.

2) Effects of the Two-Step KD: We further conduct a two-step KD with the three-VNIR-band network serving as the student and the six-VNIR-band network (OA = 0.9447) serving as the TA. Here, the parameter λ is set to 0.9, and T is set to 10.0. As shown in Table XII, with a two-step KD, the

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TABLE XI
RESULTS OF ONE-STEP KD ON THE NASC-TG2 DATASET

Student Network	λ	T	OA	
			average	std
MobileNetV3-alter-small V2, 5, 8	0	NaN	0.9087	0.0146
	0.5	1.0	0.9124	0.0084
		10.0	0.8846	0.0257
		50.0	0.9140	0.0084
	0.9	1.0	0.9128	0.0064
		10.0	0.9200	0.0182
		50.0	0.8957	0.0112
	1.0	1.0	0.9153	0.0083
		10.0	0.9212	0.0035
		50.0	0.8949	0.0143
MobileNetV3-alter-small V2, 3, 4, 5, 8, 9	0	NaN	0.9275	0.0041
	0.5	1.0	0.9213	0.0042
		10.0	0.8743	0.0312
		50.0	0.9175	0.0129
	0.9	1.0	0.9203	0.0149
		10.0	0.9333	0.0133
		50.0	0.9068	0.0142
	1.0	1.0	0.9292	0.0106
		10.0	0.9240	0.0080
		50.0	0.9138	0.0099

TABLE XII
PERFORMANCE COMPARISON FOR PROPOSED ONE- AND TWO-STEP KDS ON THE NASC-TG2 DATASET

Method	Path		OA	
			Average	std
1-step KD	Teacher: RGB OA=0.9925		0.9212	0.0035
2-step KD	Teacher: RGB OA=0.9925	TA: VNIR channels V2, 3, 4, 5, 8, 9 OA=0.9447	0.9250	0.0039

TABLE XIII
MODEL SIZE AND TIME CONSUMPTION IN INFERENCE STAGE FOR NETWORKS IN THE EXPERIMENTS WITH THE NASC-TG2 DATASET

Network & Bands	params	Inference Time (ms)
MobileNet-V3-small RGB	932.8 K	5.547
MobileNetV3-alter-small V2, 5, 8	932.8 K	5.436
MobileNetV3-alter-small V2, 3, 4, 5, 8, 9	933.2 K	5.456

performance of the student is further improved even though the knowledge is distilled between networks with distinct inputs for both teacher/TA and TA/student.

The model size and average time consumption for the abovementioned networks are shown in Table XIII. To avoid the bias caused by the capacity differences of the models, the model sizes of all the three networks are very close, and there is no significant difference in terms of the time used in the inference stage.

F. Comparison to State-of-the-Art Methods

In this section, we illustrate the performance of the proposed one- and two-step cross-modal KD strategies by comparing them with three deep learning-based algorithms trained with different strategies. For a fair comparison, all the methods employ the same backbone network and are trained on the NaSC-TG2 dataset.

TABLE XIV
OA AND STANDARD DEVIATIONS' COMPARISON WITH THE NASC-TG2 DATASET

Backbone network & Spectral bands	method	OA	
		average	std
MobileNetV3-alter-small V2, 5, 8	supervised baseline	0.9087	0.0146
	LS	0.9177	0.0056
	OLS	0.9177	0.0066
	1-step KD (proposed)	0.9212	0.0035
	2-step KD (proposed)	0.9250	0.0039
MobileNetV3-alter-small V2, 3, 4, 5, 8, 9	supervised baseline	0.9275	0.0041
	LS	0.9296	0.0161
	OLS	0.9305	0.0050
	1-step KD (proposed)	0.9333	0.0133

1) *Supervised Baseline:* The scene classification backbone network is trained in a supervised manner.
2) *Label Smoothing (LS):* As proposed in [61] and [62], the static LS paradigm utilizes soft labels generated with a uniform distribution to smooth the distribution of the hard labels.
3) *Online LS (OLS):* The nonstatic LS paradigm proposed in [63] continuously updates the soft labels during the training phase using model predictions.

During the training stage, the batch size is set to 40, and the networks are optimized with the Adam optimizer, while the learning rate is initialized to 0.0004 and decays to half for every 20 epochs. The networks are trained for 100 epochs for each method, and we repeat the experiment five times. CE losses with soft and hard labels are given equal weights in the experiments for LS and OLS methods.

As shown in Table XIV, all the methods trained with different strategies achieve better accuracy compared to the supervised baseline. Compared to the two LS methods with no distillation mechanism, the proposed one- and two-step cross-modal KD methods demonstrate better performance given input images with different spectral bands, which indicates that the proposed method is able to transfer knowledge from multimodalities during training.

Based on the experiments conducted on the NaSC-TG2 dataset, we explore the effectiveness of the cross-modal KD between classification models with distinct bands of imagery input and made the following observations. First, the supervised learning process is affected by the informativity of the input MSI and the number of available training samples. Second, the effectiveness of the distillation process is determined by the performance gap between teacher and student models. Finally, the parameters λ and T should be adjusted based on the reliability of the supervised learning and distillation process through experiments.

V. CONCLUSION

In this study, we proposed a cross-modal KD framework for RS scene classification using MSI. The core of the proposed model is to leverage the pretrained RGB image classification network to improve the performance of multispectral scene

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classification models. Moreover, to bridge the gap between modals, a two-step KD with a teach assistant network is proposed for further improvements. The evaluation of the proposed framework is conducted on the BigEarthNet and NaSC-TG2 datasets. The experimental results on the BigEarthNet dataset showed that pretrained RGB classification networks retrained for scene classification tasks can help boost the performance of student models with no pretrained weights available. The two-step KD also verified that, with a TA network, the performance of the scene classification models can be further improved. With the experiments on the NaSC-TG2 datasets, we demonstrated that KD is effective between scene classification models using imagery of totally distinct spectral bands as input, which extends the range of applications for the proposed KD framework.

For future work, we plan to investigate the usage of other definitions of knowledge for distillation in the KD framework and explore the utility of imagery from other RS sensor platforms.

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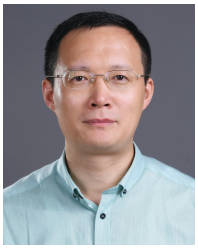
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